

MCQs:-

① B

~~can~~ solve it in the calculator because other options doesn't make any sense.

2. A

$$\begin{array}{l} \log 6 + \log 5 \\ \log(3 \times 2) + \log 5 \\ \log 3 + \log 2 + \log 5 \end{array} \quad \begin{array}{l} \log 6 + \log 5 \\ \log(6 \times 5) \\ \log(30) \end{array}$$

3. D

$$\begin{array}{l} 27m^3 + 64n^3 \\ (3m)^3 + (4n)^3 \\ (3m + 4n)((3m)^2 - (3m)(4n) + (4n)^2) \\ (3m + 4n)(9m^2 - 12mn + 16n^2) \end{array}$$

4. A

$$\begin{array}{l} 27 - 8y^3 \\ (3)^3 - (2y)^3 \\ (3 - 2y)((3)^2 + (3)(2y) + (2y)^2) \\ (3 - 2y)(9 + 6y + 4y^2) \end{array}$$

5. B

$$\begin{array}{l} |-9| - |-4| \\ 9 - 4 = 5 \end{array}$$

7) D

$$-4 \leq 2n - 2 \leq 6$$

$$-4 + 2 \leq 2n + 2 - 2 \leq 6 + 2$$

$$-2 \leq 2n \leq 8$$

$$\frac{-2}{2} \leq \frac{2n}{2} \leq \frac{8}{2}$$

$$-1 \leq n \leq 4$$

8) A

because other options doesn't make sense.

9) A

$$2a - 1 = b + 3$$

$$2(10) - 1 = b + 3$$

$$10 - 1$$

$$a = \boxed{9} \quad \boxed{5} = b$$

10) B

midpoint

$$\frac{-5 + 9}{2} = \frac{4 + (-8)}{2}$$

$$\frac{4}{2} = \frac{4 - 8}{2}$$

$$\frac{4}{2} = \frac{-4}{2}$$

$$\begin{array}{r|l} 2 & 36 \\ \hline 2 & 18 \\ \hline 2 & 9 \\ \hline 3 & 3 \\ \hline 3 & 1 \end{array}$$

Q.2(i)

(a) $\log 72$

$$\log(2^3 \times 3^2)$$

$$\therefore \log mn = \log m + \log n$$

$$\log 2^3 + \log 3^2$$

$$= 3\log 2 + 2\log 3$$

$$= 3(0.3010) + 2(0.4771)$$

$$= 0.903 + 0.9542$$

$$= 1.8572$$

(b) $\log 0.0567$

char:-

$$0.0567$$

$$5.67 \times 10^{-2}$$

~~Q.2(ii)~~

~~(a) $|x-3| + 4 = 9$~~

~~$x-3+4=9$~~

~~$x-3+4=-9$~~

Q.2(ii)

(a) $|x-3| + 4 = 9$

$$|x-3| = 9-4$$

$$|x-3| = 5$$

$$x-3 = 5$$

$$x = 5+3$$

$$x = 8$$

$$x-3 = -5$$

$$x = -5+3$$

$$x = -2$$

verification:-

$$|x-3|+4 = 9 \text{ for } x=8$$

$$|8-3|+4 = 9$$

$$|5|+4 = 9$$

$$5+4 = 9$$

$$9 = 9$$

for $x = -2$

$$|x-3|+4 = 9$$

$$|-2-3|+4 = 9$$

$$|-5|+4 = 9$$

$$5+4 = 9$$

$$9 = 9$$

$$\textcircled{b} \quad 3(n-1) \geq 2n+4$$

$$3n-3 \geq 2n+4$$

property used :- Distributive property.

$$3n-2n-3 \geq 2n-2n+4 \quad \left\{ \begin{array}{l} \text{subtrative} \\ \text{property} \end{array} \right\}$$

$$n-3 \geq +4$$

$$n \geq +3+4 \quad \left\{ \text{additive property} \right\}$$

$$n \geq 7$$

Q: 2(iii) A(-2, -1), B(1, 3), C(7, 11)

$$d_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(1 - (-2))^2 + (3 - (-1))^2}$$

$$= \sqrt{(1+2)^2 + (3+1)^2}$$

$$= \sqrt{(3)^2 + (4)^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25} \rightarrow 5$$

$$d_{BC} = \sqrt{(7-1)^2 + (11-3)^2}$$

$$= \sqrt{(6)^2 + (8)^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$= 10$$

$$d_{AC} = \sqrt{(-2-7)^2 + (-1-11)^2}$$

$$= \sqrt{(9)^2 + (12)^2}$$

$$= \sqrt{81 + 144}$$

$$= \sqrt{225}$$

$$= 15$$

Collinear or not :-

$$AB + BC = AC$$

$$5 + 10 = 15$$

$$15 = 15$$

It is having collinear points

Q:3

(i)

a) $x^4 - 8x^3 + 22x^2 - 24x + 9$
 $x^2 - 4x + 3$

x^4	$x^4 - 8x^3 + 22x^2 - 24x + 9$	x^2
x^4	$\pm x^4$	x^2
0	$-8x^3 + 22x^2$	$x^4 -$
$x(2x^2)$	$\pm 8x^3 \pm 16x^2$	
$4x^3$	$+ 6x^2 - 24x + 9$	
	$\pm 6x^2 \pm 24x \pm 9$	
	0	
	X	

square root = $\pm(x^2 - 4x + 3)^2$

b) first poly:- $x^2 - 11x + 28$

second poly:- $x^2 - 3x - 28$

HCF = ?

LCM = ?

formula:- $HCF \times LCM = P \times Q$

$\therefore HCF \times LCM = P \times Q$

$= (x^2 - 11x + 28)(x^2 - 3x - 28)$

$= (x^2 - 4x - 7x + 28)(x^2 + 4x - 7x - 28)$

$= (x(x-4) - 7(x-4))(x(x+4) - 7(x+4))$

$= (x-7)(x-4)(x+4)(x-7)$

$\therefore HCF = x-7$

$LCM = (x-4)(x+4)(x-7)$

verification :-

$$\text{HCF} \times \text{LCM} = P \times Q$$

$$(x-7) \times (x-7)(x-4)(x+4) = \frac{(x-7)(x-7)}{(x-4)(x+4)}$$

$$(x-7)^2(x^2-16) = (x-7)^2(x^2-16)$$

$$(x-7)^2(x-4)(x+4) = (x-7)^2(x-4)(x+4)$$

$$Q.3(ii) \quad P(-4, 3) \quad Q(6, -7)$$

(a) coordinates of M :-
{midpoint}

$$\left(\frac{x_1 + x_2}{2} \right), \left(\frac{y_1 + y_2}{2} \right)$$

$$\frac{-4+6}{2}, \frac{3+(-7)}{2}$$

$$\frac{+2}{2}, \frac{3-7}{2}$$

$$1, \frac{-4}{2}$$

$$(-1, -2)$$

$$P(-4, 3)$$

$$Q(6, -7)$$

$$M(1, -2)$$

$$PM = QM$$

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$$d_{PM} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(1 - (-4))^2 + (-2 - 3)^2}$$

$$= \sqrt{(1 + 4)^2 + (-5)^2}$$

$$= \sqrt{(5)^2 + 25}$$

$$= \sqrt{25 + 25}$$

$$= \sqrt{50}$$

$$= 5\sqrt{2}$$

$$d_{QM} = \sqrt{(-2 - (-7))^2 + (1 - 6)^2}$$

$$= \sqrt{(5)^2 + (-5)^2}$$

$$= \sqrt{25 + 25}$$

$$= \sqrt{50}$$

$$= \sqrt{50} \Rightarrow 5\sqrt{2}$$

$$= 5\sqrt{2}$$

$$PM = QM$$

$$5\sqrt{2} = 5\sqrt{2}$$

$$\textcircled{c} R(8, -1) \quad ? \quad M(1, -2)$$

$$= \sqrt{(1 - 8)^2 + (-2 - (-1))^2}$$

$$= \sqrt{(-7)^2 + (-2 + 1)^2}$$

$$= \sqrt{49 + 1}$$

$$= \sqrt{50}$$

$$= 5\sqrt{2}$$

$$PM = QM = RM$$

$$5\sqrt{2} = 5\sqrt{2} = 5\sqrt{2}$$

R is the same distance from M as P and Q.

$$\textcircled{a} \quad PM + MQ = PQ$$

$$\begin{aligned} dPQ &= \sqrt{(6 - (-4))^2 + (-7 - 3)^2} \\ &= \sqrt{(6 + 4)^2 + (-10)^2} \\ &= \sqrt{(10)^2 + (-10)^2} \\ &= \sqrt{100 + 100} \\ &= \sqrt{200} \\ &= 10\sqrt{2} \end{aligned}$$

$$\begin{aligned} PM + MQ &= PQ \\ 5\sqrt{2} + 5\sqrt{2} &= 10\sqrt{2} \end{aligned}$$

$PM + MQ = PQ$, M lies b/w P and Q on the straight line segment PQ.

P, M, Q are collinear, with M b/w P and Q.